# Ilan Sela

Texas A&M University
Dept. of Electrical & Computer Engineering

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## RESEARCH INTERESTS

Generative modeling and theoretical machine learning

## EDUCATION

**Texas A&M University** — College Station, TX
M.S. in Computer Engineering — Expected: May 2027
Advisor: Dr. Tie Liu
*GPA: 4.0/4.0*

**Texas A&M University** — College Station, TX
B.S. with honors in Electrical Engineering — 2022 – 2025
Minor in Computer Science
*GPA: 4.0/4.0*

## RESEARCH EXPERIENCE

**Texas A&M University** — College Station, TX

***Graduate Researcher (with Prof. Tie Liu)*** — May 2025 – Present

- Developing a theoretical understanding of one-step image generators by studying drift-based generative training dynamics.

***Graduate Researcher (with Prof. Dileep Kalathil)*** — May 2026 – October 2026

- Reduced reliance on fixed, human-designed search rules by developing AgentDiscover, in which a coding agent directs scientific discovery and proposes candidates.
- Improved long-horizon search efficiency by designing a graph-based memory that allows fresh agents to retrieve and reuse prior ideas and candidate lineages.

***Research Project*** — January 2026 – May 2026

*Reinforcement Learning-Based Calibration for Radar Systems*

- Reduced reliance on costly measurement equipment and repeated radar access by developing an offline reinforcement learning framework for automated MIMO calibration.
- Improved reconstructed image quality by formulating calibration as a Markov decision process with a target-independent image-quality reward.

***Undergraduate Researcher (with Prof. Tie Liu)*** — August 2024 – May 2025

- Addressed query-free black-box optimization by fine-tuning FLAN-T5 to generate candidate designs directly from offline data.
- Compared one-step and multi-step generation strategies, producing designs that exceeded the best offline examples.

***Undergraduate Researcher*** — May 2023 – May 2024

Secure and Trustworthy Hardware (SETH) Lab

- Integrated and validated a reinforcement learning-based query-selection policy within the lab's existing FuncTeller black-box hardware reconstruction attack.

## PUBLICATIONS AND RESEARCH OUTPUT

- Mahdi Farahbakhsh, **Ilan Sela**, Fatemeh Doudi, Vishnu Teja Kunde, Krishna Narayanan, Jean-Francois Chamberland, Dileep Kalathil. "AgentDiscover: Autonomous Discovery with Minimal Search Scaffolding." *arXiv:2610.05334*, 2026. *Under review.*
- Sumin Kim, **Ilan Sela**, Muhammad Faeyz Karim (2026). A Reinforcement Learning-Based Self-Calibration Framework for Radar Systems in Offshore Crane Applications. Presentation, 2026 Mary Kay O'Connor Safety & Risk Conference
- **Ilan Sela** (2025). Black-Box Offline Optimization via Large Language Models. Undergraduate Research Scholars Program. Available electronically from https://hdl.handle.net/1969.1/1591266

## ACADEMIC HONORS

**Graduate**

- Qualcomm Graduate Scholarship Recipient (2025 – 2026)

**Undergraduate**

- Undergraduate Research Scholar, Engineering Honors, Summa Cum Laude, Dean's Honor Roll (2022 – 2025)

TEACHING EXPERIENCE

**Agentic Intelligence for Discovery and Exploration (AIDE) Workshop mentor**, Texas A&M University August 2026

- Mentored faculty and graduate students in designing and implementing agentic workflows for research problems.

**Student Assistant** for "ECEN 740: Machine Learning Engineering", Texas A&M University January 2025 – May 2025

INDUSTRY EXPERIENCE

**Luminator Technology Group** Plano, TX

*Embedded Software and Electrical Engineering* Intern May 2025 – August 2025

- Supported Flying Whales aircraft systems by developing embedded firmware for onboard microcontrollers.
- Eliminated cabin rewiring by developing a wireless LED-control retrofit for CRJ aircraft; the system became the company's primary R&D focus.
- Enabled LED driving and CAN-bus distribution for aerospace lighting systems through PCB designs in Altium.

TECHNICAL SKILLS

**Languages:** Python, C/C++, CUDA, MATLAB, Verilog, Java, Assembly

**ML/Scientific Computing:** PyTorch, TensorFlow, NumPy, SciPy

**Embedded/Hardware:** STM32, PIC, Altium

SELECTED COURSEWORK

- Machine Learning, Reinforcement Learning, Online Learning, Information Theory, Estimation and Detection Theory, Convex Optimization, GPU Programming